CMAgent: A Multi-Agent System for Automated Cell Type–Marker Curation from Literature

Abstract

We present CMAgent, a multi-agent system for automated cell type–marker curation from biomedical literature. CMAgent integrates literature retrieval and prioritization, evidence-sentence classification with LLM-refined training labels, and downstream multi-agent curation with verification and normalization. LLM-refined labels improved sentence-classification F1 from 0.43 to 0.69, while the BiomedBERT ensemble achieved 0.84 top-10 precision compared with 0.55 for the MarkerGeneBERT baseline. Downstream curation achieved an F1 of 0.52, while verification reduced false positives by 5.96% without recall loss. Standardized records retain supporting evidence and are available through a searchable web portal.
Keywords: Cell marker, Biomedical text classification, Biomedical relation extraction, Multi-agent system

1. Introduction

Cell type–marker relationships are essential for annotating and interpreting single-cell transcriptomic data, and resources such as CellMarker provide manually curated marker knowledge [1]. However, the rapid growth of biomedical literature makes it increasingly difficult to keep such resources up to date through manual curation alone, motivating automated literature-based curation.

Prior approaches such as MarkerGeneBERT automate marker-related sentence classification and cell type–marker extraction from biomedical literature [2]. Although these approaches substantially reduce manual effort, reliable identification of marker-related sentences remains challenging, while relation extraction and entity standardization also remain difficult for large-scale curation.

To address these challenges, we developed CMAgent, a multi-agent system for automatically identifying, extracting, and standardizing cell type–marker relationships from biomedical literature. CMAgent combines LLM-refined evidence sentence selection with multi-agent relation curation to produce standardized records linked to supporting literature and accessible through a searchable web portal.

2. Methodology

CMAgent processes biomedical literature through literature retrieval and prioritization, sentence segmentation and evidence-sentence classification, and multi-agent cell type-marker relation curation, followed by storage of standardized records in a searchable database and web portal (Fig. 1).

2.1. Evidence-Sentence Selection

Candidate biomedical literature is retrieved from PubMed, with available full text or abstracts segmented into sentences. Candidate papers are then prioritized by semantic similarity to CellMarker references using SPECTER2, a scientific-document embedding model [3].

For evidence-sentence classification, the original marker-related sentence labels from MarkerGeneBERT [2] are independently reassessed by GPT-5.5 and Claude Opus. Sentences on which the two LLMs agree provide refined labels for retraining BiomedBERT classifiers [4]. An ensemble of the retrained classifiers provides final evidence scoring. Selected evidence sentences are supplemented with species and tissue context before downstream relation curation.

2.2. Multi-Agent Relation Curation

The selected evidence sentences are processed by a Controller-guided multi-agent framework for relation extraction, risk-gated verification, and identifier normalization. The Controller identifies known cell-type and marker names in each sentence and provides them as hints to a zero-shot Extraction Agent, which extracts relations supported by the text. Rather than verifying every relation, the Controller sends only relations expressed in potentially indirect or ambiguous contexts to a Verification Agent. Extracted cell types and markers are first matched against locally stored Cell Ontology and NCBI Gene mappings. When no match is found, separate Normalization Agents select the most appropriate identifier from retrieved Cell Ontology or NCBI Gene candidates. All downstream LLM-based agents use GPT-4o-mini.

3. Results

3.1. Evidence-Sentence Selection

We first evaluated SPECTER2-based literature prioritization, using CellMarker references published through 2022 as reference papers and post-2022 CellMarker references as held-out evaluation papers. SPECTER2 retained 4,114 of 4,149 held-out references within the top 60% of a 468,386-paper PubMed corpus, corresponding to 99.16% reference recovery. The resulting 281,032 papers were used to construct the production corpus of 33.6M

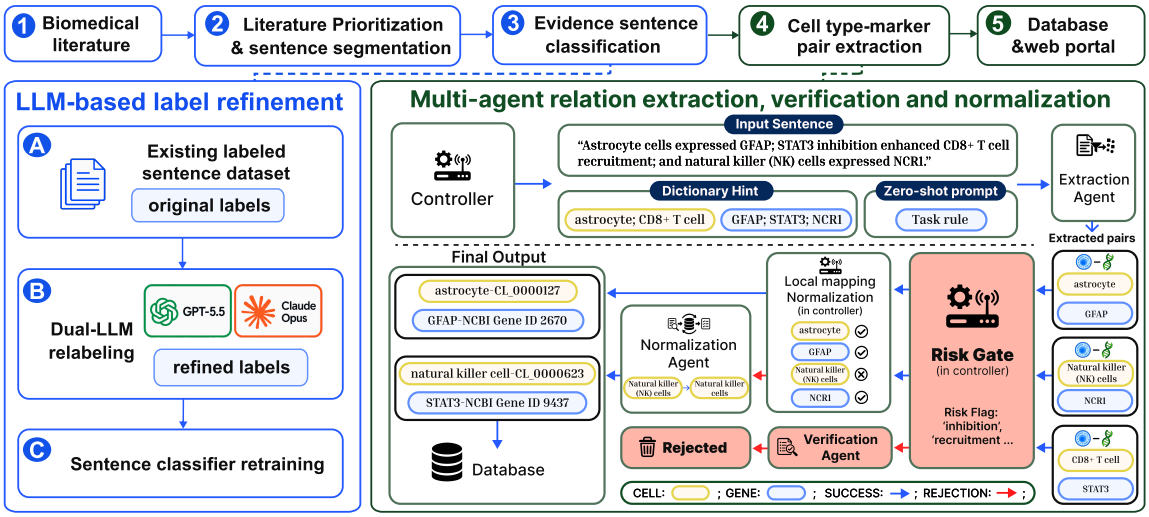


Figure 1: Overview of the proposed framework.

segmented sentences.

We next evaluated the effect of label refinement on sentence classification and assessed generalization on 57 CellMarker-supported papers not used for training. For the external evaluation, the top 50 candidate sentences from each paper were human-labeled, and each model ranked the candidate set using top-10 precision, recall, and F1.

Training with refined labels improved sentence-classification F1 from 0.43 to 0.69. In the external evaluation, the BiomedBERT ensemble achieved 0.84 top-10 precision, compared with 0.55 for the MarkerGeneBERT spaCy baseline [5] (Table 1). A probability threshold selected using 39 papers and evaluated on 18 held-out papers yielded 0.78 precision, 0.88 recall, and 0.83 F1. Applying the final classifier to the production corpus selected 2.73M (8.13%) evidence-sentence candidates for downstream cell type–marker relation curation.

Table 1: Top-10 evidence-sentence ranking performance.

Model	Labels	P@10	R@10	F1@10
spaCy (MarkerGeneBERT)	Original	0.55	0.23	0.33
spaCy	Refined	0.66	0.28	0.39
BiomedBERT	Original	0.61	0.26	0.36
BiomedBERT	Refined	0.82	0.35	0.49
BiomedBERT ensemble	**Refined**	**0.84**	**0.36**	**0.50**

3.2. Downstream Curation and Web Portal

We evaluated downstream cell type–marker curation on a human-labeled benchmark of 376 PubMed/PMC sentences from 87 PMIDs, comprising 743 normalized CellMarker relations. Predictions were matched within sentences by NCBI Gene ID and either specific or broad CL ID.

Downstream curation achieved an F1 of 0.52, with all ablations reducing F1 (Table 2). Notably, 88.73% of sentence/PMID-specific false positives were curated under other PMIDs in CellMarker. Risk-gated verification reduced false positives by 5.96% without recall loss, while normalization resolved 7.26% of locally unmapped cell mentions and 52.94% of gene mentions. CMAgent also exceeded MarkerGeneBERT NER in pre-normalization string-level F1 (0.46 vs. 0.35).

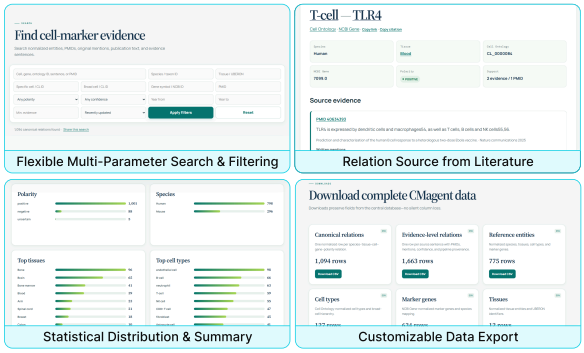


Figure 2: Major features of the CMAgent web portal.

The curated records are accessible through the CMAgent web portal for search, evidence inspection, summary, and export (Fig. 2).

Table 2: Ablation results of CMAgent.

Method	F1	ΔF1
CMAgent	**0.52**	–
w/o Dictionary Hint	0.49	-0.03
w/o Verification Agent	0.51	-0.01
w/o Normalization Agent	0.51	-0.01

4. Conclusions

CMAgent links LLM-refined evidence selection with structured multi-agent curation to build traceable cell type–marker records provided through a searchable web portal. Evaluation is limited by PMID-specific CellMarker gold coverage; future work will broaden expert validation and corpus coverage.

References

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